



SILVER OAK UNIVERSITY
Silver Oak College of Computer Application
Master of Computer Application
Course Name: Computer Network
Course Code: CAM5021C
Semester:1st

Prerequisite:

Possess basic knowledge of computer systems, operating systems and number systems. Familiarity with fundamental data communication concepts and common Internet usage is desirable.

Course Objectives:

This course provides a comprehensive understanding of computer networking, covering fundamentals, communication methods, IP addressing, routing, and standard network models. It also equips students with essential knowledge of network security principles, threats, and protective measures.

Teaching Scheme:

Teaching & Learning Scheme (In Hrs Per Semester)									Total Credits
Classroom Instruction						Extended Learning		Total Hours	
L		T		P					
Cr	Hr	Cr	Hr	Cr	Hr	TW	SL	Hr	C
4	60	0	0	0	0	40	20	60	4

Contents:

Unit	Topics	Teaching Hours	% Weightage
1	Basic of Network: Basic Network, Needs of Computer Networks, History of Computer Network, Networking Models, Centralized, Decentralized, Peer-to- Peer, Client/Server. Types of Computer Networks: LAN, MAN, WAN, PAN, SAN Network Topologies : Definition, Types of Topology, Advantages, Disadvantages Network Media & Devices: Transmission Media : Guided, Unguided Hub, Switch, Repeater, Router, Bridge, Firewall, Gateway	17	28
2	Communication Using A Network: Data Communication, Signalling Techniques, Principal Of data communication, Data transferring techniques IP Addressing & Routing: Introduction of ipv4, Need of IP addressing, Subnetting, Network services, Introduction of ipv6, Configuration of ipv4 & ipv6, Difference: ipv4, ipv6, Routing Protocols, Algorithms	14	24
3	OSI Model & TCP/IP Model: Introduction about OSI Model, Understanding the Purpose and Function of Networking Models, Layered Architecture, OSI Layer and its Applications. TCP/IP Model: Network Access Layer, Internet Layer, Host-to-Host Layer, Application Layer, ASCII code.	11	18

4	Intelligent Network Security: Introduction, Definition & Objective: CIA triad, AAA, Threats & Vulnerabilities, Cryptography fundamentals, Common Attacks, Security Monitoring and Future Trends in Intelligent Network Security.	18	30
Total		60 Hours	100%

Course Outcomes (For Theory):

CO No.	CO Statement	Blooms Taxonomy Level	Unit
CO-1	Explain the fundamentals of computer networks, networking models, network types, topologies, transmission media, and networking devices along with their functions and applications.	2	1
CO-2	Apply data communication principles, signaling techniques, IP addressing, subnetting methods, and routing protocols to configure and manage computer networks.	3	2
CO-3	Analyze the layered architecture, functions, and protocols of the OSI and TCP/IP models to evaluate their role in network communication.	4	3
CO-4	Evaluate network security requirements, threats, vulnerabilities, cryptographic techniques, security monitoring mechanisms, and emerging intelligent network security technologies.	5	4

Course Outcomes (For Practical):NA

CO No.	CO Statement	Blooms Taxonomy Level	Unit	Practical Cluster

Teaching & Learning Pedagogy approaches:

1. Lectures: Provide foundational knowledge and conceptual understanding.
2. Assignments: Regular assignments to reinforce learning and application.
3. Case Studies.

List of Tutorials: NA

Sr. No.	Tutorial Content	Hours	Co No.
1		1	
2		1	
3		1	
Total Hours			

List of Experiments: NA

Sr. No.	Tutorial Content	Hours	Co No.
1		1	
2		1	
3		1	
Total Hours			

Major Equipment and Simulation Tools:

1. Computer System
2. Switches and HUB
3. LAN cable, Crimping Tools
4. Cisco Packet Tracer simulation tool
5. Network Simulator-3 simulation tool

Books Recommended:

1. Kurose and Ross, "Computer Networking- A Top-Down approach", Pearson
2. Behrouz Forouzan, "Computer Networks- A Top-Down approach", McGraw Hill
3. Andrew Tanenbaum, "Computer Networks", Prentice Hall

Term Work Module:

40 HOURS

Objective:

To strengthen conceptual understanding and practical skills in Computer Networks through AI-assisted design, analysis, simulation, and security activities.

Term Work	Statement	Hours	CO No.
Term Work 1	Design a Smart Campus Network using AI tools (ChatGPT/Gemini/Copilot). Select appropriate topology, transmission media, and networking devices. Prepare a network design report with AI-assisted recommendations.	10	1
Term Work 2	Perform subnetting, IPv4/IPv6 addressing, and routing exercises. Use AI tools to generate network scenarios and analyze routing decisions and traffic optimization techniques.	10	2
Term Work 3	Analyze how data is transmitted between two computers through different layers of the OSI and TCP/IP models. Observe packet encapsulation, decapsulation, and protocol behavior using network simulation tools.	10	3
Term Work 4	Security Assessment of a Smart Campus Network.	10	4
Total Hours		40	

Self-Learning Module:**20 HOURS****Objective:** To encourage independent learning of emerging trends in networking.**Recommended Online Resources:**

Sr. No.	Unit Name/Topic Name	Resource Link
1	Network Simulator Tool	Search Tutorials spoken-tutorial.org
2	Networking Basics	https://www.netacad.com
3	Intelligent Network Security	https://learn.microsoft.com/security/ https://www.ibm.com/topics/network-security

CO-PO-PSO Matrix (For Theory):

Co. No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2
CO-1	3	1	-	-	-	1	2	2	2
CO-2	3	3	1	-	2	2	2	3	2
CO-3	3	3	-	-	-	-	2	2	2
CO-4	2	3	3	1	2	2	2	2	3

CO-PO-PSO Matrix (For Practical): NA

Co. No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PSO1	PSO2
CO-1									
CO-2									
CO-3									
CO-4									

CO-PO-PSO MAPPING JUSTIFICATION FOR Computer Network (for Theory)

CO No.	CO Statement	PO No.	Justification
CO1	Explain the fundamentals of computer networks, networking models, network types, topologies, transmission media, and networking devices along with their functions and applications.	PO1	Students develop a strong understanding of basic networking concepts and principles that form the foundation of computer science knowledge.
		PO2	Students learn to identify networking components and understand their roles in communication systems.

		PO6	Students explain networking concepts, technologies, and their applications using appropriate technical terminology.
		PSO1	The knowledge of networking fundamentals supports the development and understanding of computer-based systems and applications.
		PSO2	Students gain awareness of networking technologies that are widely used in the computing industry.
CO2	Apply data communication principles, signaling techniques, IP addressing, subnetting methods, and routing protocols to configure and manage computer networks.	PO1	Students apply fundamental networking and computing concepts to perform network configuration and management tasks.
		PO2	Students analyze networking requirements and select suitable addressing, subnetting, and routing techniques to solve networking problems.
		PO3	Students choose appropriate networking solutions based on the requirements of different communication scenarios.
		PO5	Students develop skills related to planning, configuring, and managing network resources effectively.
		PSO1	Students use networking concepts and techniques to support the design and operation of computer-based applications and systems.
		PSO2	Students gain practical exposure to networking technologies commonly used in professional environments.
CO3	Analyze the layered architecture, functions, and protocols of the OSI and TCP/IP models to evaluate their role in network communication.	PO1	Students build a thorough understanding of network architectures and communication protocols.
		PO2	Students examine protocol operations and analyze how data is transmitted across different network layers.
		PO6	Students interpret and explain the interactions among various network layers and protocols.

		PSO1	Understanding protocol architectures helps students work with network-enabled applications and systems.
		PSO2	Students become familiar with standard networking framework and communication technologies used in the industry.
CO4	Evaluate network security requirements, threats, vulnerabilities, cryptographic techniques, security monitoring mechanisms, and emerging intelligent network security technologies.	PO1	Students apply security principles and computing knowledge to understand and protect networked systems.
		PO2	Students identify security issues, analyze vulnerabilities, and assess potential risks in network environments.
		PO3	Students evaluate different security approaches and recommend suitable solutions for securing networks.
		PO4	Students consider security challenges that often involve knowledge from multiple domains such as networking, cryptography, and intelligent systems.
		PO5	Students understand the importance of implementing, monitoring, and managing security measures within organizations.
		PO6	Students communicate security concerns, risks, and mitigation strategies using appropriate technical methods.
		PO7	Students stay updated with evolving cybersecurity threats, tools, and technologies through continuous learning.
		PSO1	Knowledge of network security helps students build and maintain secure computer-based applications and systems.
		PSO2	Students gain exposure to current and emerging security technologies that support professional growth in the computing field.

CO-PO-PSO MAPPING JUSTIFICATION FOR Computer Network (for Practical): NA

CO No.	CO Statement	PO No.	Justification

MAPPING WITH SUSTAINABLE DEVELOPMENT GOALS

SDG Mapping for Computer Network (CAM5021C)		
SDG No. and Title	Relevance to Course	Justification
SDG 4: Quality Education	High	Computer Networks provide foundational and advanced knowledge required for digital learning, communication, and technology-driven education.
SDG 9: Industry, Innovation and Infrastructure	High	Networking technologies form the backbone of modern digital infrastructure and support innovation in communication systems and intelligent networks.
SDG 16: Peace, Justice and Strong Institutions	Moderate	Network security and threat detection contribute to secure digital environments and information protection.

Explanation:

The Computer Network course contributes to Quality Education by enhancing digital and technical competencies. It supports Industry, Innovation and Infrastructure through networking technologies, intelligent networking solutions. The course also promotes secure and responsible use of digital infrastructure through cybersecurity and network security practices.